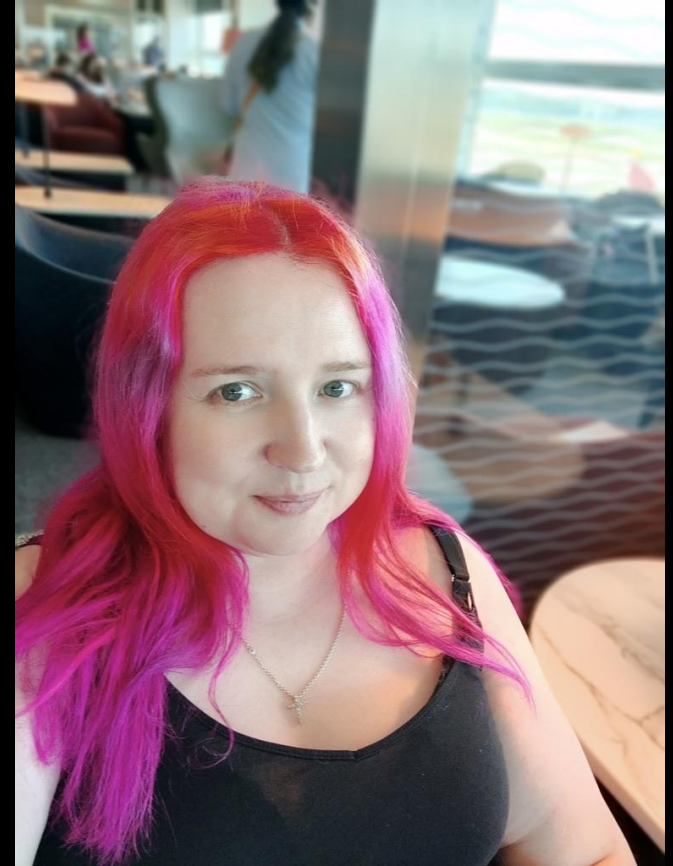


Observing your PHP application with OpenTelemetry

Jessica Smith, July 2025

About Me

- PHP user since 1998
- Originally from Gloucester, now in Reading
- Senior Principal Software Engineer for DigiCert
- Senior Developer at Fasthosts in Gloucester
- LAN Party Gamer



What is Observability?



What is Observability?

Gaining **understanding** of a system by observing and measuring its outputs.

Why do we want this?

- Detecting Problems
- Quicker and easier troubleshooting
- Identifying Bottlenecks

Logs

- PSR-3
- Monolog
- Centralised – Logstash, Splunk

```
[2025-07-04 10:30:47] production.DEBUG: [Party:1] SLAN: Received simple webhook event
[2025-07-04 10:30:47] production.DEBUG: [Party:1] SLAN: Updating from PartyUpdate job
[2025-07-04 10:30:47] production.INFO: [Party:1] SLAN: Updating state
[2025-07-04 10:30:47] production.DEBUG: [Party:1] SLAN: Updating current song
[2025-07-04 10:30:47] production.DEBUG: [User:1] Mintopia: Spotify API -> getMyCurrentPlaybackInfo()
[2025-07-04 10:30:47] production.DEBUG: [User:1] Mintopia: Creating new API connection
[2025-07-04 10:30:47] production.INFO: [Party:1] SLAN: Updating current song to [Song:3374] Drop It Like It's Hot
[2025-07-04 10:30:47] production.DEBUG: [Party:1] SLAN: Spotify API -> getTrack(2NBQmPr0EEjA8Vbew0QGx0)
[2025-07-04 10:30:48] production.DEBUG: [Party:1] SLAN Updating history
[2025-07-04 10:30:48] production.DEBUG: [User:1] Mintopia: Spotify API -> getPlaylist(3pHPgF7cgkZfdn1uJbGGGm)
[2025-07-04 10:30:48] production.DEBUG: [Party:1] SLAN: Spotify API -> deletePlaylistTracks(3pHPgF7cgkZfdn1uJbGGGm, [])
[2025-07-04 10:30:48] production.DEBUG: [User:1] Mintopia: Spotify API -> getPlaylist(3pHPgF7cgkZfdn1uJbGGGm)
[2025-07-04 10:30:48] production.DEBUG: [Party:1] SLAN: Updating playlist
[2025-07-04 10:30:48] production.DEBUG: [Party:1] SLAN: Spotify API -> addPlaylistTracks(3pHPgF7cgkZfdn1uJbGGGm, [])
[2025-07-04 10:30:48] production.INFO: [Party:1] SLAN: Adding [Song:3442] 4 Chords to queue
[2025-07-04 10:30:48] production.DEBUG: [Party:1] SLAN: Spotify API -> queue(1cu5PHumWAqQf9M6BHQ8b7, [])
```

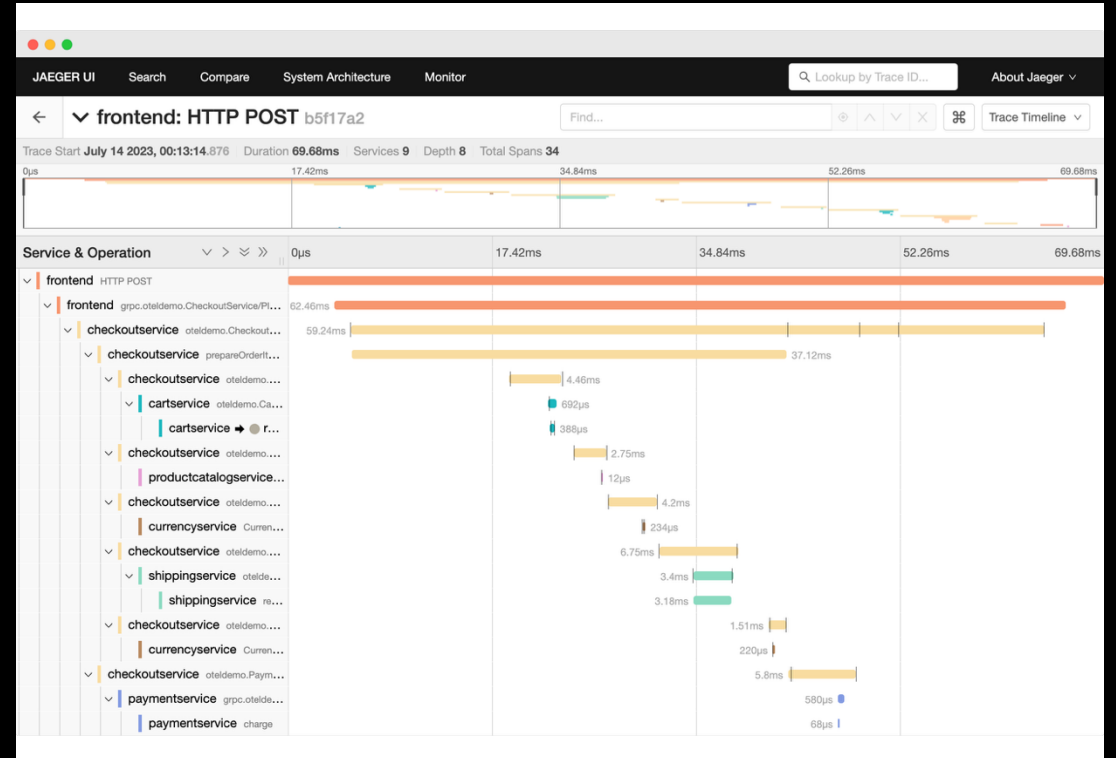
Metrics

- Key Performance Indicators
- Many different metrics:
 - Response Time
 - Exceptions
 - Successful/Failed Logins
 - So many things ...
- Prometheus



Traces

- Following request from start to finish
- If using microservices – across microservices
- Measure performance of operations (spans) within a trace



Bringing it all together

- Three Pillars
 - Logs – The textual narrative
 - Metrics – Quantitative data
 - Traces – Visualising the flow of a request
- Combined, we get a **holistic** view with **context**

Observability Tools and Services

Self-Hosted

- Jaegar
- Uptrace
- OpenObserve
- SigNoz

Services

- Honeycomb
- Datadog
- New Relic
- Splunk
- AWS Cloud Watch
- So many...

How do we implement it?

- Platform and vendor agnostic
- Minimal effort
- Some sort of **standard**

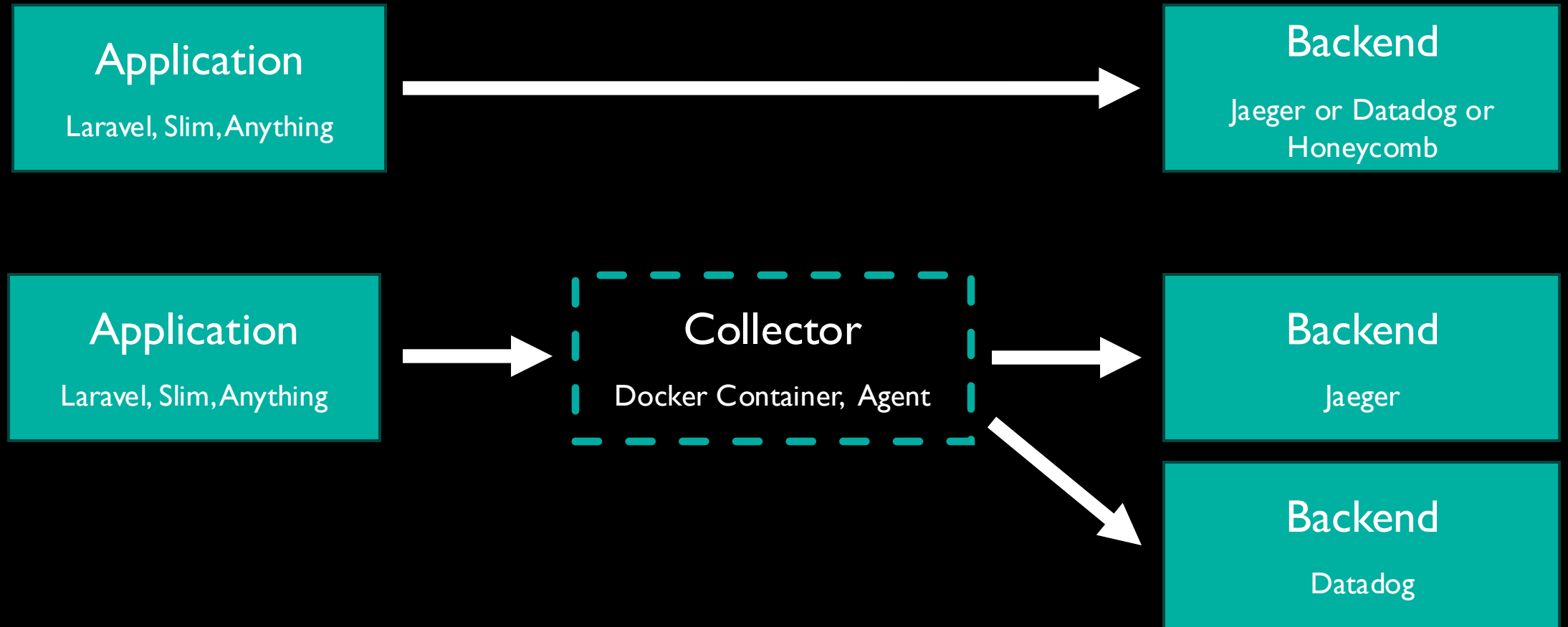
I have a suggestion...

OpenTelemetry

- Open Source
- CNCF Incubating Project
- Vendor Neutral
- Well Supported
- Zero-code required



How it works



Collector

- Agent or Docker Container
- YAML Config
- Defines pipelines for data



- Extensions and Filters

Zero Code?

1. PHP Extension
2. Composer Packages
3. Environment Variables

PHP Extension

- Based on zend_observer
- Allows hooks on function and method calls
- Install through PECL, php-extension-installer or PIE



```
pie install open-telemetry/ext-opentelemetry
```


Composer Packages

- Uses the hooks from the extension
- Support for many different frameworks, PSRs, packages

```
composer require open-telemetry/sdk
```

```
composer require open-telemetry/opentelemetry-auto-Laravel
```

```
composer require open-telemetry/opentelemetry-auto-psr15
```

Environment Variables

Set using docker run or DotEnv

```
OTEL_SERVICE_NAME=application
```

```
OTEL_EXPORTER_OTLP_ENDPOINT=http://collector:4318
```

```
OTEL_EXPORTER_OTLP_PROTOCOL=http/json
```

Demo

Let's do this live!

Going further

- Manual Instrumentation
- Use attribute based hooks
- Write your own hooks for your application
- Export some metrics
- Sampling
 - You don't want or need all the data

```
OTEL_TRACES_SAMPLER=traceidratio  
OTEL_TRACES_SAMPLER_ARG=0.2
```

Resources

- OpenTelemetry - <https://opentelemetry.io>
- PHP Installer for Extensions - <https://github.com/php/pie>
- Slides - <https://github.com/mintopia/php-opentelemetry-talk-2025>

Thank you

<https://phpc.social/@mintopia>

<https://github.com/mintopia>

<https://discord.gg/roave>

Slides

